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Assignment	Analysis of Algorithm
Dept	AI 4-1
Date	03/05/26

Problem :- 1 :- Quadratic :-

The formula of sum of arithmatic Progression is :

$$S = N/2 \, (2a_1 + (N-1)d)$$

Rearranging

$$dN^2 + (2a_1 - d)N - 2S = 0$$

$$N = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

By solving we will solve by pseudo code and implementing the algorithm.

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Algorithm in Psudo code.

{

Input first term a_1, d, S

$a = d$, $b = (2a_1 - d)$, $c = -2S$

Calculate discriminent

$D = b^2 - 4ac$

if $D<0$ No solution

else

$N = (-b + \sqrt{D})/2a$

$Y = a_1 + (N-1)d$

Print D & Y.

Psudocode:

{

Input a_1, d, S

$a = d$

$b = 2a_1 - d$

$c = -2S$

$D = b^2 - 4ac$

If $D<0$

Print "No real solution";

Else

$N = (-b + \sqrt{D})/2a)$

$Y = a_1 + (N-1)d$

Print N

Prin Y

END.

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C++ Program:

```
#Include <iostream>
#Include <cmath>
Using namespace std;

int main() {
    double a1, d, S;
    double a, b, c, D, N, Y;

    cout<<"Enter first term: ";
    cin>> a1;
    cout<< " Enter d";
    cin >> d;
    cout << " Enter sum";
    cin>> S;
    a= d
    b= 2*a1 - d;
    c= -2*S;

    if (D<0) {
        cout <<" No real solution";
    }else {
        N= (-b+ sqrt (D)) /(2*a);
        Y= a1 + (N-1)*d;

        cout<<" No of (N): "<<N<<endl;
        cout<< last term (Y): <<Y<<endl;
    }
    return 0.
}
```

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Time complexity:-

Best : O(1)

Average : O(1)

Worst : O(1)

Problem 2:- Factorization:-

Lets Start the algorithm

Input a_1, d, S

sum = 0 , N = 0

while sum < S

term = a1 + Nd

sum = sum + term

N = N + 1

Last term = $a_1 + (N-1)d$

Show N & last term.

Psuodo Code:-

Start

Input a_1, d, S

sum = 0

N = 0

While sum < S

term = a_1 + N*d

sum = sum + term

N = N + 1

End while

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$Y = a1 + (N-1)*d$

Print N
Print Y
END.

C++ Program:-

```
#Include <iostream>
using namespace std;
int main() {
    double a1, d, S;
    double sum = 0
    int N = 0;
    double term, Y;

  cout<< "Enter first :" ;
   cin >> a1;
  cout << "Enter (d) :" ;
   cin >> d;

while (sum < S) {
      term = a1 + N*d
      sum = sum + term
      N++ ;

}
  Y = a1 + (N-1)*d;
```

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```
cout<<" No of term " <<N <<endl;
cout<< " Last term <<Y <<endl;

return 0;
}
```

Time complexity:-

Best case: (O)1 (if S is small)
Average : O(N)
Wost : O(N)

Problem :: 3 : Comparison.

Feature	Quadratic	Factorization
Time complex.	O(1)	O(N)
Speed	Fast	slow
Use of loops	No	Yes
Efficiecy	High	medium
Use	larger valu	small valuey

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Problem :4 :- Genelarization

Both algorithms work for any values of

First term a1
common diff (d)
Sum (s)

we take input from user.

overall Quadratic method is more efficiant and faster because it uses formula.

x.———x.